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Homo- and hetero-polymerization of ethylene catalyzed by α-diimine nickel complexes with hydroxyl groups

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ABSTRACT

Heterogeneous olefin polymerization catalysts have received wide interests from academia and industry because their distinct advantages in thermostability, avoidance of reactor fouling and continuous production operations. In this work, hydroxy functional groups are introduced to the α-diimine nickel framework, which can serve as reactive functionality to react with $MgCl_2/AlR_n(OEt)_{3-n}$ and form ionic anchoring interaction that bind the catalyst strongly to the surface of the supports. These heterogeneous catalysts are highly active in ethylene polymerization to produce polyethylene with higher molecular weight (up to 280.5×10^4 g·mol^{-1}), lower branching density and higher tensile strength (up to 26.7 MPa) than those generated by the homogeneous counterparts. More importantly, the nickel complexes with multiple hydroxy groups lead to high loading capacity on $MgCl_2/AlR_n(OEt)_{3-n}$ supports, affording polyethylene products with minimum inorganic contamination.

1. Introduction

The milestone discovery of Brookhart-type α-diimine palladium and nickel complexes in 1995 has promoted the revolution of olefin polymerization field [1]. Recently, various homogeneous high-performance catalysts based on late transition metals have been reported for olefin (co)polymerization process [2–12]. In polyolefin industry, heterogeneous catalysts stand out attributed to their distinct advantages in stability at high polymerization temperature, avoidance of reactor fouling along with continuous production operations [13–15]. In addition, heterogeneous polymerization was highly controllable to produce polyethylenes with well-defined product morphological [16–19]. The design and development of efficient heterogeneous catalytic systems applicable to industrial polymerization processes have received wide interests from both academia and industry [20–23].

The heterogenization strategy of metal catalysts on the surface of solid supports has been widely studied, and the combination of solid supports or cocatalyst-modified supports with metal complexes was achieved by hydrogen bond, covalent bond, or ionic bond (Scheme 1) [24–26]. For example, Cai and coworkers reported the synthesis of heterogeneous naphthoquinone-based nickel and palladium catalysts through hydrogen bonding interactions of the ligands with the SiO_2 surface, which can produce high-molecular-weight polar-functionalized semicrystalline polyethylene along with great morphology control (Scheme 1, **I**) [27,28]. Recently, polymers such as polystyrene (PS) and polymethyl methacrylate (PMMA) could also be used as supports for heterogenization, generating polyolefins products with minimum inorganic contamination in polymerization. For example, heterogeneous catalysts based on cross-linked PS via covalent bond were studied by Mecking and coworkers (Scheme 1, **II**) [29]. Emulsion polymerization strategy was introduced to prepared various imino-based nickel olefin polymerization catalysts supported by cross-linked PMMA, improving

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the mechanical and surface properties of polyolefin products (Scheme 1, **III**) [30]. Ionic bond is another common method for anchoring catalyst to supports and has been widely investigated. A versatile ionic anchoring strategies were exploited sufficiently in different catalytic systems to provide heterogeneous catalyst for producing polar polyolefins, and different inorganic substrates (SiO_2, ZnO, Al_2O_3, TiO_2, MgO) were investigated (Scheme 1, **IV**) [31,32]. In another example, trifluoroborate potassium substituents were installed on an α-diimine framework, which endowed the α-diimine nickel complex with both strong affinity for SiO_2 supports and electron-donating nature for the enhancement of stability and molecular weight of polyethylene products (Scheme 1, **V**) [33]. Imine-basic catalysts bearing sodium-sulfonate ionic groups were synthesized and immobilized on SiO_2 support to investigate their performance in ethylene (co)polymerization, offering free-flowing polyethylene with excellent morphology control (Scheme 1, **VI**) [34].

The research of catalyst-support interaction is fascinating because it provides "drop-in" catalyst solutions for existing technologies in the current polyolefin industry [35,36]. However, there are still some issues left to be solved for potential industrialization of heterogeneous late transition metal catalysts, including low catalytic activity and inorganic impurities in polyethylene products [37]. In this contribution, a serious of α-diimine nickel complexes with multiple hydroxyl substituents were synthesized and supported on $MgCl_2/AlR_n(OEt)_{3-n}$ through ionic bonding. These heterogeneous catalysts were employed in ethylene polymerization to produce polyethylene with great mechanical properties (See Scheme 2).

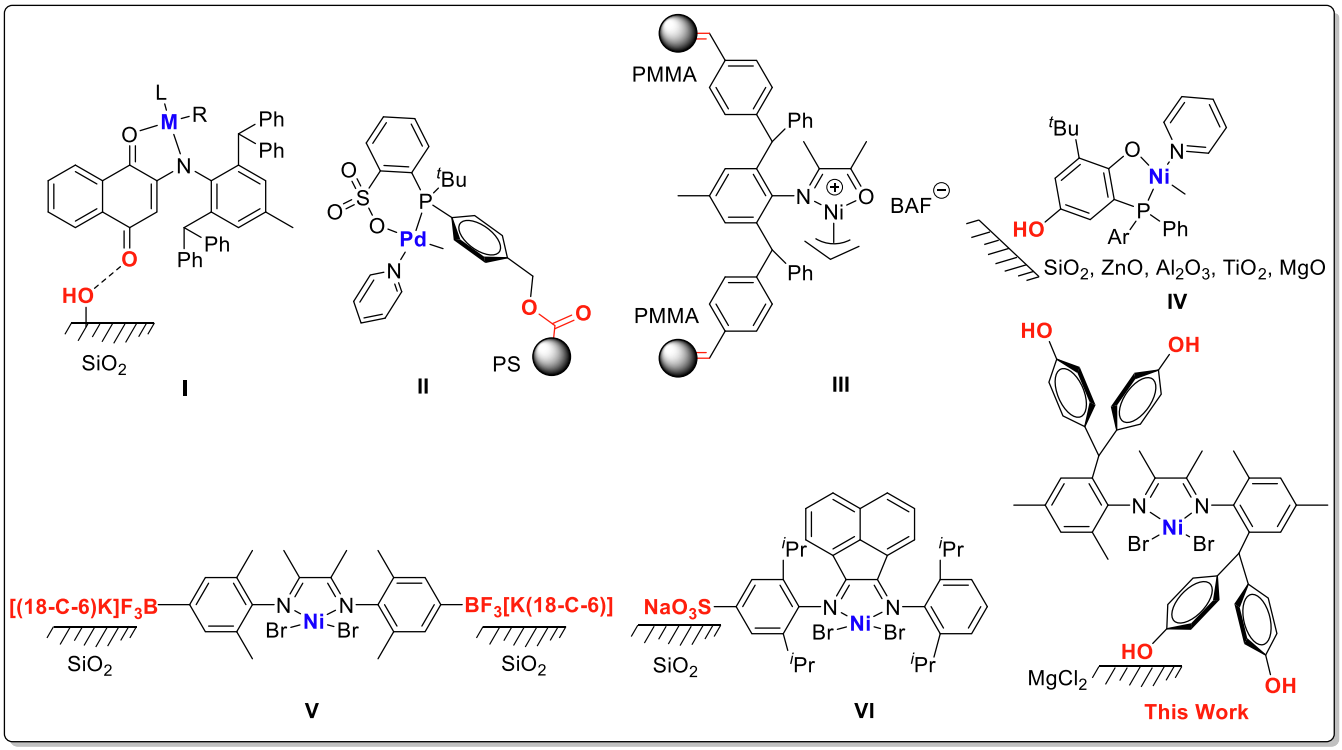
2. Results and discussion

2.1. Synthesis and characterization of the catalysts

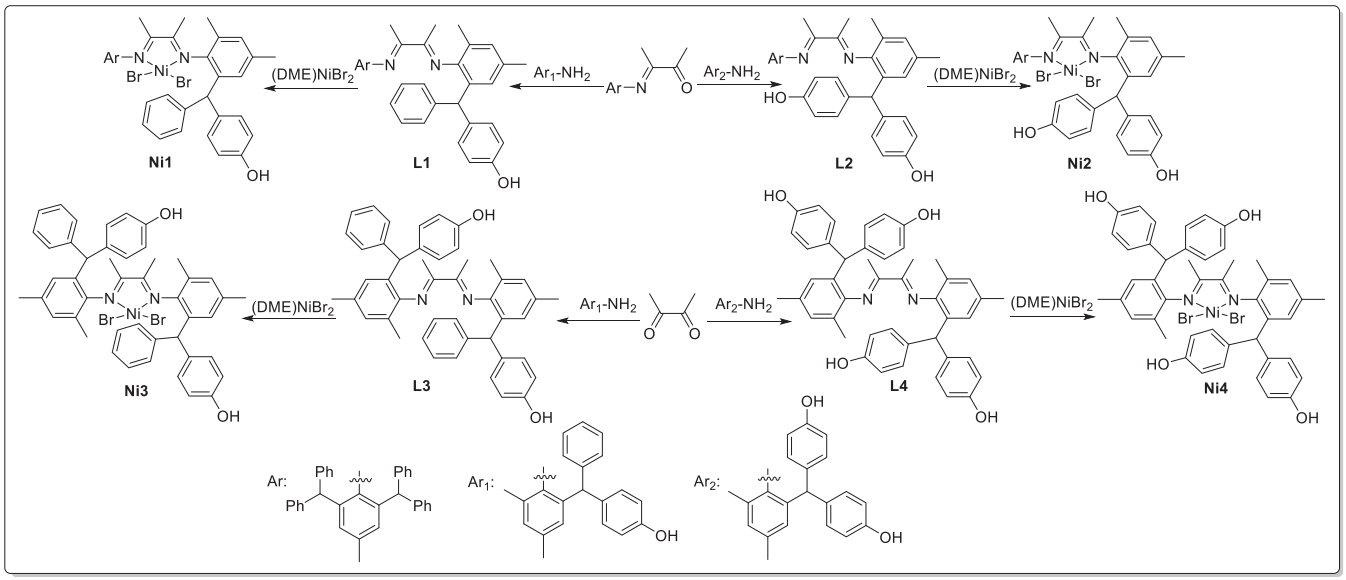
2-((4-Methoxyphenyl)(phenyl)methyl)-4,6-dimethylaniline and 2-(bis(4-methoxyphenyl)methyl)-4,6-dimethylaniline were synthesized according to the literature procedures [38,39]. $MgCl_2/AlR_n(OEt)_{3-n}$ was prepared as the literature-reported procedures [40]. The aniline bearing hydroxyl substituents (Ar_1-NH_2, Ar_2-NH_2) were prepared with excess BBr_3 in 83–95 % yields. The ligands (**L1**-**L4**) were synthesized by the acid-catalyzed condensation reactions and characterized by ^1H and ^{13}C NMR analyses. The corresponding Ni complexes (**Ni1**-**Ni4**) were prepared in excellent yields (80–90%) by coordination reactions of the ligands with one equivalent of $(DME)NiBr_2$ (DME = 1,2-dimethoxyethane) for 12 h. These nickel complexes were characterized by mass spectrometry and elemental analysis. The molecular structure of **Ni1** complex was determined by X-ray diffraction (Fig. 1). The nickel center adopted a distorted tetrahedron geometry, which is similar to previously reported α-diimine nickel complexes [41,42].

2.2. Homogeneous polymerization

Firstly, different alkyl aluminum compounds were used as co-catalyst to study the catalytic performance, including Et_2AlCl and MMAO (Table 1, entries 1–2). Considering the catalytic activities and economic benefits, Et_2AlCl was chosen as co-catalyst in the following ethylene polymerization. In the presence of Et_2AlCl cocatalyst, all of these nickel complexes (**Ni1**-**Ni4**) are active in ethylene polymerization (level of 10^6 g·mol^{-1}·h^{-1}; Table 1). The dependence of polymerization activity on temperature was not observed with all nickel complexes in this system, possibly due to the balance of catalyst stability, mass transfer effects and ethylene solubility with temperature. At 50 °C, **Ni1** generated polyethylene with high branching density (94/1000C) and high molecular weight (85.1 × 10^4 g·mol^{-1}; Table 1, entry 1). As the polymerization temperature increased at 80 °C, the molecular weight of polyethylene products was reduced obviously (45.4 × 10^4 g·mol^{-1}; Table 1, entry 3), which was consistent with the literature and ascribed to the increased rate of β-H elimination [43,44]. Under the same condition, complexes **Ni2** led to comparable branching density and molecular weight (Table 1, entries 4–5). Compare to **Ni1** and **Ni2**, complexes **Ni3** and **Ni4** led to the formation of polyethylene products with much lower branching density and higher molecular weight (Table 1, entries 6–11). It may be due to two factors. First, the quasi-centrosymmetric structures of **Ni3** and **Ni4** may contribute to low rate



Scheme 1. Selected Heterogeneous catalysts for producing polyolefins.



Scheme 2. Syntheses of ligands and nickel complexes. DME = 1,2-dimethoxyethane.

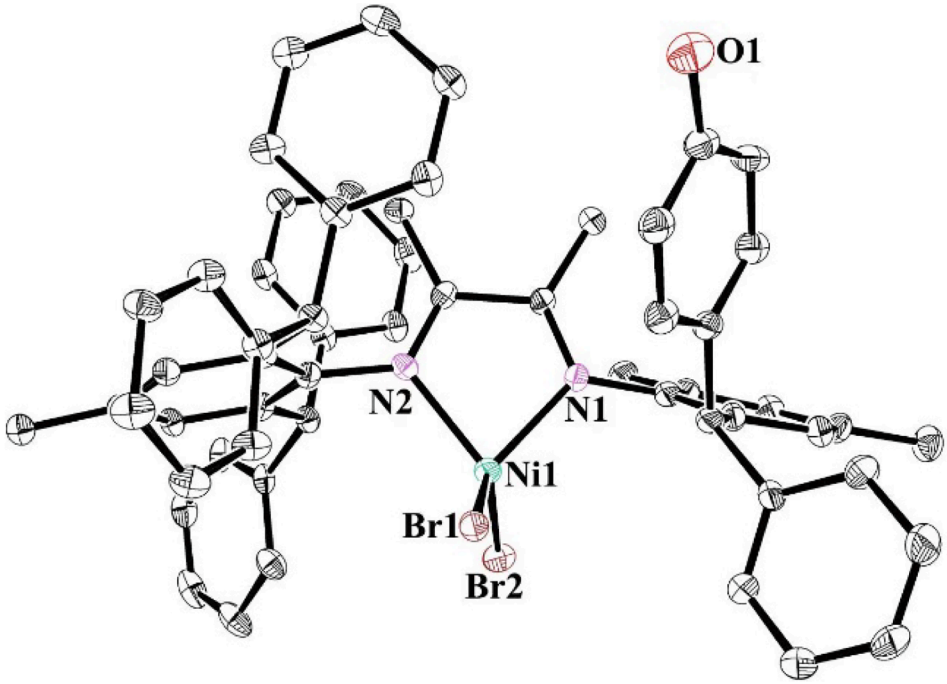


Fig. 1. X-ray crystallographic structures of **Ni1** (CCDC Number, 2297980). Selected bond lengths (Å) and angles (°): Ni1-N1 1.997(6), Ni1-N2 2.005(6), Ni1-Br2 2.3573(15), Ni1-Br1 2.3258(14), N1-Ni1-N2 81.4(2), Br1-Ni1-Br2 124.90(6), N1-Ni1-Br1 119.31(18), N1-Ni1-Br2 101.18(18), N2-Ni1-Br1 104.74(17), N2-Ni1-Br2 117.63(17).

of chain walking [45,46]. Second, the electron-donating effect of the hydroxyl groups may increase the electron density on the nickel center and slows down the chain transfer reaction [47].

2.3. Heterogeneous polymerization

It was reported that α-diimine nickel complexes can be easily immobilized on $MgCl_2/AlR_n(OEt)_{3-n}$ by alkyl aluminum activating the α-diimine nickel complexes to form polymerization sites and ion pairs that bind the catalysts to the supports surface [48]. In this work, hydroxy functional groups were introduced to the ligand framework, which can react with alkyl aluminum in $MgCl_2/AlR_n(OEt)_{3-n}$ supports to provide strong affinity via ionic bond interaction. And our experimental results demonstrated that the larger amount of hydroxy groups on the ligand may lead to greater loading capacity, probably due to the fact that hydroxy groups were reactive functionality, which react with $MgCl_2/AlR_n(OEt)_{3-n}$ supports. The maximum loading of **Ni1**-**Ni4** were determined by inductively coupled plasma (ICP) analysis and followed the order: **Ni4** (ca. 7 μmol/100 mg) > **Ni3**(ca. 4 μmol/100 mg) ≈ **Ni2**(ca. 4 μmol/100 mg) > **Ni1**(ca. 3 μmol/100 mg). The excellent loading capacity of these nickel complexes were much higher than those observed in previous reports [48,49], providing benefit of minimum inorganic contamination in the as-prepared polyethylene product finally.

The catalytic activities of these $MgCl_2$-supported nickel catalysts

Table 1
Polymerization of ethylene catalyzed by **Ni1-Ni4**.[a)]

Ent.	Cat.	T (°C)	Yield (g)	Act. (10^6)[b)]	B[c)]	T_g/T_m (°C)[d)]	M_n (10^4)[e)]	*PDI*[e)]	*Strain (%)*[f)]	*Stress (MPa)*[f)]	*SR (%)*[g)]
1	**Ni1**	50	1.0	12.0	94	−55/4	85.1	4.5	410 ± 53.5	1.6 ± 0.3	90
2[h)]	**Ni1**	50	0.4	4.8	56	−40/–	72.4	2.0	400 ± 29.6	5.2 ± 0.6	68
3	**Ni1**	80	0.9	10.8	98	−57/–9	45.4	5.7	–	–	–
4	**Ni2**	50	1.0	12.0	83	−53/10	84.2	3.5	420 ± 26.4	1.9 ± 0.1	86
5	**Ni2**	80	0.9	10.8	94	−58/–6	53.1	4.1	–	–	–
6	**Ni3**	50	1.5	18.0	29	−46/78	150.6	2.2	470 ± 9.3	12.4 ± 0.1	50
7	**Ni3**	80	1.4	16.8	70	−56/–	58.4	2.2	500 ± 39.2	4.5 ± 0.2	73
8[i)]	**Ni3**	100	1.0	6.0	80	−59/–	43.9	2.1	–	–	–
9	**Ni4**	50	0.6	7.2	26	–/103	174.6	2.1	420 ± 9.0	18.9 ± 0.7	31
10	**Ni4**	80	0.9	10.8	53	−51/70	47.5	2.9	810 ± 39.1	22.4 ± 1.8	46
11[i)]	**Ni4**	100	0.8	4.8	79	−50/–	22.3	3.3	–	–	–

[a)] Conditions: Ni catalyst 0.5 μmol, hep = 18 mL, CH_2Cl_2 = 2 mL, ethylene = 8 atm, time = 10 min, Et_2AlCl = 500 eq. [b)] Activity is in unit of 10^6 g·mol^{-1}·h^{-1}. [c)] Branches per 1000 carbon atoms, determined by ^{1}H NMR in $C_2D_2Cl_4$ at 120 °C. [d)] Determined by differential scanning calorimetry (DSC). [e)] Determined by gel permeation chromatography (GPC) in trichlorobenzene at 150 °C with polystyrene standards. M_n is in unit of 10^4 g·mol^{-1}. [f)] Strain and stress at break values. [g)] The strain recovery values (SR) can be calculated by SR = 100 ($\varepsilon_a - \varepsilon_r$)/$\varepsilon_a$, where ε_a is the applied strain and ε_r is the strain in the cycle at zero load after 10th cycle. [h)] MMAO = 500 eq. [i)] Time = 20 min.

were slightly lower than their homogeneous counterparts in ethylene polymerization (Table 2), which may be due to the complications caused by spatial site crowding and side reactions in heterogeneous system [16]. Under the same condition, heterogeneous catalysts resulted in polyethene products with higher molecular weights and lower branching density than their homogeneous catalyst analogs in most experiments (Table 2 versus Table 1), which may be contributed to the increased steric environment around the nickel center and the decreased chain transfer rate induced by heterogenization [33]. More importantly, the highest molecular weight was obtained by **Ni4@MgCl$_2$** at 50 °C (Table 2, entry 8, up to 280.5 × 10^4 g·mol^{-1}), which exceeded some heterogeneous systems in the literature reports [33,34]. In addition, low glass transition temperature value (T_g) was observed in homogeneous polymerization, while their heterogeneous counterparts prepared polymer without T_g in most experiments, which may be because that polymer produced by heterogeneous catalysts possessed higher molecular weight and lower branching density (Table 2). And much higher melting temperature (T_m) were observed of all polyethylene products produced by $MgCl_2$-supported nickel catalysts (113–125 °C). It may due to the interaction of the hydroxy moiety on the ligand backbone with the $MgCl_2/AlR_n(OEt)_{3-n}$ supports alters the steric/electronic environment around the nickel center, influencing the polymerization properties [50]. Furthermore, the original data of ^{13}C NMR spectra of the polymers generated by **Ni1** and **Ni1@MgCl$_2$** at 50 °C were employed (Figs. S1 and S2), the branching types of both polymers included methyl branch mainly. The ethylene polymerization was employed for a prolonged time using complexes **Ni3** and **Ni3@MgCl$_2$** at 100 °C to study the thermal stability (Fig. 2a). Compared to **Ni3**, heterogeneous **Ni3@MgCl$_2$** remained its high performance up to 1 h. This phenomenon suggested that heterogeneous **Ni3@MgCl$_2$** possessed much better thermostability than homogeneous **Ni3**.

2.4. Properties of polyolefin products

Stress/strain experiments were performed at standard room temperature (25 °C) to investigate mechanical and material properties of the polyethylene products obtained in this system. The high branching density and low crystallinity of polymers generated by homogeneous **Ni1** and **Ni2** at 50 °C translate into soft and elastic materials with low tensile strength (1.6 MPa and 1.9 MPa) and high strain at break values (410 % and 420 %; Fig. 2b), as well as very high elastic recovery values (ca. 90 % and 86 %; Table 1, entries 1 and 4; Fig. 2e). These as-prepared ploymers using ethylene as the only feedstock exhibited both mechanical and elastic properties, which can be used as EPOEs materials [49,51]. As the polymerization temperature increased to 80 °C, the polyethylene products prepared by **Ni1** and **Ni2** showed poor mechanical properties, probably due to the combine of low molecular weight and high branching density (Table 1, entries 3 and 5). Compare to homogeneous **Ni1** and **Ni2**, at 50 °C, heterogeneous **Ni1@MgCl$_2$** and **Ni2@MgCl$_2$** afford polymers with higher tensile strength (6.1 and 17.5 MPa; Fig. 2b) and lower elastic recovery values (ca. 70 % and 47 %; Table 2, entries 1 and 3), which may originate from the differences in polymers topological structure (molecular weight and branching density) and the reinforcing effect from the dispersed inorganic

Table 2
Polymerization of ethylene catalyzed by heterogeneous **Ni1-Ni4**.[a)]

Ent.	Cat.	T (°C)	Yield (g)	Act (10^6)[b)]	B[c)]	T_g/T_m (°C)[d)]	M_n (10^4)[e)]	*PDI*[e)]	*Strain (%)*[f)]	*Stress (MPa)*[f)]	*SR (%)*[g)]
1	**Ni1@MgCl$_2$**	50	0.7	8.4	75	−61/113	168.3	2.2	400 ± 7.2	6.1 ± 0.4	70
2	**Ni1@MgCl$_2$**	80	0.8	9.6	95	−60/118	96.9	4.3	660 ± 43.7	3.4 ± 0.2	74
3	**Ni2@MgCl$_2$**	50	0.6	1.8	54	–/121	104.4	1.7	330 ± 18.7	17.5 ± 1.1	47
4	**Ni2@MgCl$_2$**	80	0.8	2.4	92	−59/120	80.2	4.8	340 ± 4.9	3.7 ± 0.5	64
5	**Ni3@MgCl$_2$**	50	1.2	14.4	30	–/117	158.4	3.9	420 ± 27.1	19.0 ± 1.5	28
6	**Ni3@MgCl$_2$**	80	1.4	16.8	57	–/116	68.9	4.1	630 ± 46.3	11.4 ± 1.1	50
7[h)]	**Ni3@MgCl$_2$**	100	1.4	8.4	60	–/119	47.6	3.5	770 ± 25.2	14.4 ± 0.9	47
8	**Ni4@MgCl$_2$**	50	0.9	2.7	21	–/125	280.5	1.5	630 ± 23.7	26.7 ± 0.6	26
9	**Ni4@MgCl$_2$**	80	1.6	4.8	39	–/123	117.0	2.6	660 ± 41.9	20.4 ± 1.5	35
10[h)]	**Ni4@MgCl$_2$**	100	1.3	2.0	57	–/124	86.6	4.7	980 ± 13.2	16.2 ± 1.1	42
11[h)]	**Ni4@MgCl$_2$**	120	1.3	2.0	92	−61/124	37.2	4.9	1210 ± 59.3	4.4 ± 0.1	45

[a)] Conditions: Ni catalyst 0.5 μmol, hep = 20 mL, ethylene = 8 atm, time = 10 min, Et_2AlCl = 500 eq. [b)] Activity is in unit of 10^6 g·mol^{-1}·h^{-1}. [c)] Branches per 1000 carbon atoms, determined by ^{1}H NMR in $C_2D_2Cl_4$ at 120 °C. [d)] Determined by differential scanning calorimetry (DSC). [e)] Determined by gel permeation chromatography (GPC) in trichlorobenzene at 150 °C with polystyrene standards. M_n is in unit of 10^4 g·mol^{-1}. [f)] Strain and stress at break values. [g)] The strain recovery values (SR) can be calculated by SR = 100 ($\varepsilon_a - \varepsilon_r$)/$\varepsilon_a$, where ε_a is the applied strain and ε_r is the strain in the cycle at zero load after 10th cycle. [h)] Time = 20 min.

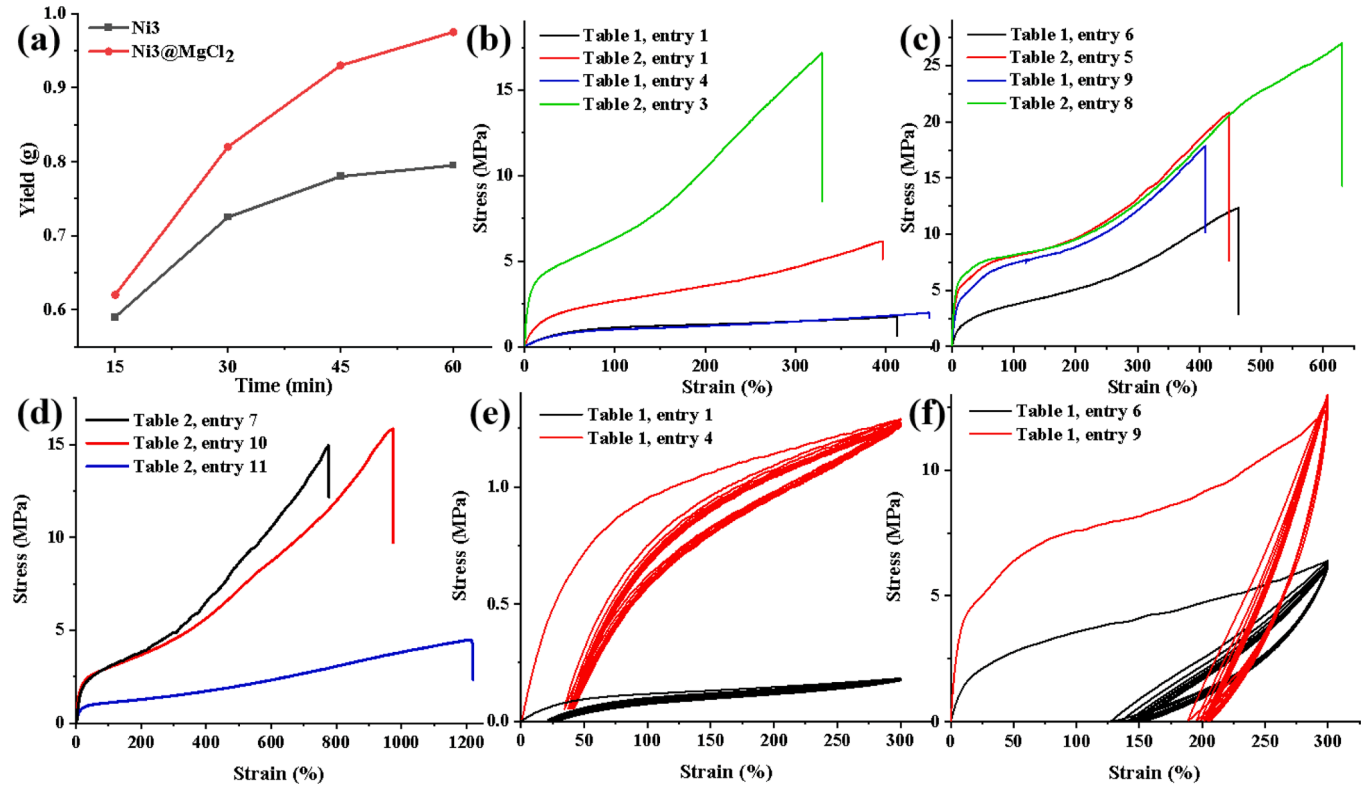


Fig. 2. (a) Plots of polymer yield versus polymerization time for homogeneous catalysts **Ni3** and heterogeneous **Ni3@MgCl$_2$** at 100 °C. (b) Stress strain curves of the polyethylene products prepared from homogeneous catalysts **Ni1**/**Ni2** and heterogeneous **Ni1@MgCl$_2$**/**Ni2@MgCl$_2$** at 50 °C. (c) Stress strain curves of the polyethylene products prepared from homogeneous catalysts **Ni3**/**Ni4** and heterogeneous **Ni3@MgCl$_2$**/**Ni4@MgCl$_2$** at 50 °C. (d) Stress strain curves of the polyethylene products prepared from heterogeneous **Ni3@MgCl$_2$** at 100 °C and **Ni4@MgCl$_2$** at 100 °C and 120 °C. (e) Stress–strain curve of the polymer samples (Table 1, entries 1 and 4) in the hysteresis experiments for a strain of 300 %. (f) Stress–strain curve of the polymer samples (Table 1, entries 6 and 9) in the hysteresis experiments for a strain of 300 %.

nanoparticles [49,52]. At 80 °C, complexes **Ni1@MgCl$_2$** and **Ni2@MgCl$_2$** still maintained the capability of generating super stretchable polymer product, although the tensile stress was reduced (Table 2, entries 2 and 4).

The polymers generated by **Ni3** and **Ni4** at 50 °C all showed poor elastic recovery values (ca. 50 % and 31 %; Table 1, entries 6 and 9; Fig. 2f), while the tensile strength was high (12.4 and 18.9 MPa; Fig. 2c). This probably due to the high molecular weight and low branching density. At an elevated temperature (80 °C), complexes **Ni3** and **Ni4** still able to produce polymers with excellent mechanical properties (Table 1, entries 7 and 10). Comparable tensile strength and increased strain values were observed of the polymers produced by heterogeneous **Ni3@MgCl$_2$** and **Ni4@MgCl$_2$** at 50 °C (Table 2, entries 5 and 8; Fig. 2c). Furthermore, the heterogeneous complex **Ni3@MgCl$_2$** and **Ni4@MgCl$_2$** can afford polyethylene products with excellent mechanical and material properties even at 100 °C and 120 °C (Table 2, entries 7 and 10–11, Fig. 2d).

3. Conclusion

In summary, a series of α-diimine nickel complexes with reactive hydroxy functional groups were synthesize and immobilized on $MgCl_2$/$AlR_n(OEt)_{3-n}$ supports to prepared heterogeneous catalysts. The homogeneous nickel complexes are highly active in ethylene polymerization. Among them, complexes **Ni1** and **Ni2** generated polyethylenes with both mechanical and elastic properties. Complexes **Ni3** and **Ni4** led to the formation of polyethylene products with much lower branching density and higher molecular weights due to their symmetrical structure. The heterogeneous catalyst analogs exhibited great thermal stability, producing polyethylenes with higher molecular weight (up to 280.5×10^4 g·mol^{-1}) and higher tensile strength (up to 26.7 MPa) than those generated by the homogeneous counterparts. The complex **Ni4** with multiple hydroxy groups led to greater loading capacity on $MgCl_2$/$AlR_n(OEt)_{3-n}$ supports, affording polyethylene products with minimum inorganic contamination finally. The study of these hydroxyl-substituted α-diimine nickel complexes in this work will inspire the design of heterogeneous catalysts.

4. Author statement

† Nan Nie and Fan Yu conducted the experiments and contributed equally. Dan Peng and Binyuan Liu conceived the idea, designed the experiments and evaluated the data. Dan Peng and Binyuan Liu wrote the manuscript together. Dan Peng acquired the financial funding support for this project.

CRediT authorship contribution statement

Nan Nie: Data curation. **Fan Yu:** Investigation. **Binyuan Liu:** Formal analysis. **Dan Peng:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

The Supporting Information is available free of charge on the Elsevier Publications website. Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.112962.

References

[1] L.K. Johnson, C.M. Killian, M. Brookhart, New Pd(II)- and Ni(II)-based catalysts for polymerization of ethylene and α-olefins, J. Am. Chem. Soc. 117 (23) (1995) 6414–6415.
[2] H. Deng, H. Zheng, H. Gao, L. Pei, H. Gao, Late transition metal catalysts with chelating amines for olefin polymerization, Catalysts 12 (9) (2022) 936.
[3] R. Wu, W.K. Wu, L. Stieglitz, S. Gaan, B. Rieger, M. Heuberger, Recent advances on α-diimine Ni and Pd complexes for catalyzed ethylene (co)polymerization: a comprehensive review, Coord. Chem. Rev. 474 (2023) 214844.
[4] S. Mecking, M. Schnitte, Neutral nickel(II) catalysts: from hyperbranched oligomers to nanocrystal-based materials, Acc. Chem. Res. 53 (11) (2020) 2738–2752.
[5] H. Mu, G. Zhou, X. Hu, Z. Jian, Recent advances in nickel mediated copolymerization of olefin with polar monomers, Coord. Chem. Rev. 435 (2021) 213802.
[6] H. Mu, L. Pan, D. Song, Y. Li, Neutral nickel catalysts for olefin homo- and copolymerization: relationships between catalyst structures and catalytic properties, Chem. Rev. 115 (22) (2015) 12091–12137.
[7] F. Lin, T. Morgen, S. Mecking, Living aqueous microemulsion polymerization of ethylene with robust Ni(II) phosphinophenolato catalysts, J. Am. Chem. Soc. 143 (49) (2021) 20605–20608.
[8] S.M.M. Mortazavi, H. Jafarian, M. Ahmadi, S. Ahmadjo, Characteristics of linear/branched polyethylene reactor blends synthesized by metallocene/late transitional metal hybrid catalysts, J. Therm. Anal. Calorim. 123 (2016) 1469–1478.
[9] Y. Maddah, S. Ahmadjo, S.M.M. Mortazavi, F. Sharif, D. Hassanian-Moghaddam, M. Ahmadi, Control over branching topology by introducing a dual catalytic system in coordinative chain transfer polymerization of olefins, Macromolecules 53 (11) (2020) 4312–4322.
[10] M. Omidvar, S. Mansouri, S.M.M. Mortazavi, S. Ahmadjo, Shedding light on the poly(5-ethylidene-2-norbornene) microstructural differences: Skeleton rearrangements during polymerization, J. Appl. Polym. Sci. 141 (2) (2024) e54775.
[11] S. Mansouri, M. Omidvar, S.M.M. Mortazavi, S. Ahmadjo, 5-Ethylidene-2-norbornene polymerization by α-diimine nickel catalyst: a revealing insight into the pivotal function of binuclear and mononuclear catalyst structure in tailoring polymer architecture, Macromol. React. Eng. 16 (2) (2022) 2100052.
[12] A. Talaei, F.A. Taromi, A. Arefazar, S. Ahmadjo, O.M. Jazanic, The influence of bridge type on the activity of supported metallocene catalysts in ethylene polymerization, Chinese J. Polym. Sci. 32 (2) (2014) 137–142.
[13] A.R. Albunia, F. Prades, D. Jeremic, Multimodal Polymers with Supported Catalysts: Design and Production, Springer, Switzerland, 2019.
[14] P. Preishuber-Pflugl, M. Brookhart, Highly active supported nickel diamine catalysts for polymerization of ethylene, Macromolecules 35 (16) (2002) 6074–6076.
[15] H.S. Schrekker, V. Kotov, P. Preishuber-Pflugl, P. White, M. Brookhart, Efficient slurry-phase homopolymerization of ethylene to branched polyethylenes using α-diimine nickel(ii) catalysts covalently linked to silica supports, Macromolecules 39 (19) (2006) 6341–6354.
[16] C. Zou, C. Tan, C. Chen, Heterogenization strategies for nickel catalyzed synthesis of polyolefins and composites, Acc. Mater. Res. 4 (6) (2023) 496–506.
[17] T. Liang, S. Goudari, C. Chen, A simple and versatile nickel platform for the generation of branched high molecular weight polyolefins, Nat. Commun. 11 (1) (2020) 372.
[18] H. Tafazolian, D.B. Culver, M.P. Conley, A well-defined Ni(II) α-diimine catalyst supported on sulfated zirconia for polymerization catalysis, Organometallics 36 (13) (2017) 2385–2388.
[19] L.C. Simon, H. Patel, J.B.P. Soares, R.F. de Souza, Polyethylene made with in situ supported Ni-diimine/SMAO: replication phenomenon and effect of polymerization conditions on polymer microstructure and morphology, Macromol. Chem. Phys. 202 (17) (2001) 3237–3247.
[20] M.M. Stalzer, M. Delferro, T.J. Marks, Supported single-site organometallic catalysts for the synthesis of high-performance polyolefins, Catal. Lett. 145 (2015) 3–14.
[21] M.C. Baier, M.A. Zuideveld, S. Mecking, Post-metallocenes in the industrial production of polyolefins, Angew. Chem. Int. Ed. 53 (37) (2014) 9722–9744.
[22] B.C. Damien, W.D. Rick, V. Amrit, M. Jittima, J.R. Aaron, A.P. Evgeny, S. L. Andrew, R.L. Graham, P.C. Matthew, Active sites in a heterogeneous organometallic catalyst for the polymerization of ethylene, ACS Cent. Sci. 7 (7) (2021) 1225–1231.
[23] M. Khoshsefat, S. Ahmadjo, S.M.M. Mortazavi, G.H. Zohuri, Reinforcement effects of nanocarbons on catalyst behaviour and polyethylene properties through *in situ* polymerization, RSC Adv. 6 (91) (2016) 88625–88632.
[24] T.R. Boussie, C. Coutard, H. Turner, V. Murphy, T.S. Powers, Solidphase synthesis and encoding strategies for olefin polymerization catalyst libraries, Angew. Chem. Int. Ed. 37 (23) (1998) 3272–3275.
[25] C. Favero, D.W. Lima, R.F. Souza, K. Bernardo-Gusmão, Ethylene polymerization with a nickel diimine catalyst covalently anchored on spherical ZSM-5, New J. Chem. 41 (6) (2017) 2333–2339.
[26] M. Okada, Y. Nakayama, T. Shiono, Heterogenization of an anphthoquinone-chelated nickel complex for ethylene polymerization using silica-supported modified methylaluminoxane, Macromol. Chem. Phys. 215 (18) (2014) 1792–1796.
[27] F. Xia, L. Zhang, R. Tanaka, T. Shiono, Z. Cai, Highly robust nickel catalysts containing anilinonaphthoquinone ligand for copolymerization of ethylene and polar monomers, Macromolecules 50 (23) (2017) 9216–9221.
[28] H. Zhang, C. Zou, H. Zhao, Z. Cai, C. Chen, Hydrogen-bonding-induced heterogenization of nickel and palladium catalysts for copolymerization of ethylene with polar monomers, Angew. Chem. Int. Ed. 133 (32) (2021) 17586–17591.
[29] P. Wucher, J.B. Schwaderer, S. Mecking, Solid-supported single-component Pd (II) catalysts for polar monomer insertion copolymerization, ACS Catal. 4 (8) (2014) 2672–2679.
[30] D. Peng, M. Xu, C. Tan, C. Chen, Emulsion polymerization strategy for heterogenization of olefin polymerization catalysts, Macromolecules 56 (6) (2023) 2388–2396.
[31] C. Zou, G. Si, C. Chen, A general strategy for heterogenizing olefin polymerization catalysts and the synthesis of polyolefins and composites, Nat. Commun. 13 (1) (2022) 1954.
[32] C. Zou, Q. Wang, G. Si, C. Chen, A co-anchoring strategy for the synthesis of polar bimodal polyethylene, Nat. Commun. 14 (1) (2023) 1442.
[33] C. Hong, Z. Wang, H. Jiang, G. Si, M. Song, C. Chen, Dual roles of trifluoroborate in nickel-catalyzed ethylene polymerization: electronic perturbation and anchoring for heterogenization, Chinese Chem. Lett. 34 (8) (2023) 107918.
[34] Y. Sun, Q. Wang, Y. Pan, W. Pang, C. Zou, M. Chen, SiO2-supported Ni(II) and Fe (II) catalysts bearing sodium- sulfonate group for olefin polymerization, Chinese J. Chem. 40 (23) (2022) 2773–2779.
[35] G.G. Hlatky, Heterogeneous single-site catalysts for olefin polymerization, Chem. Rev. 100 (4) (2000) 1347–1376.
[36] A. Dorcier, N. Merle, M. Taoufik, F. Bayard, C. Lucas, A. de Mallmann, J.M. Basset, Preparation of a well-defined silica-supported nickel-diimine alkyl complex-application for the gas-phase polymerization of ethylene, Organometallics 28 (7) (2009) 2173–2178.
[37] D.B. Malpass, Introduction of industrial polyethylene: properties, catalysts, and processes, (2010).
[38] B. Wang, H. Liu, C. Zhang, T. Tang, X. Zhang, Propylene homopolymerization and copolymerization with ethylene by acenaphthene-based α-diimine nickel complexes to access EPR-like elastomers, Polym. Chem. 12 (43) (2021) 6307–6318.
[39] S. Dai, G. Li, W. Lu, Y. Liao, W. Fan, Suppression of chain transfer via a restricted rotation effect of dibenzosuberyl substituents in polymerization catalysis, Polym. Chem. 12 (22) (2021) 3240–3249.
[40] Y. Nakayama, J. Saito, H. Bando, T. Fujita, Propylene polymerization behavior of fluorinated bis(phenoxy-imine) Ti complexes with an MgCl2-based compound (MgCl2-supported Ti-based catalysts), Macromol. Chem. Phys. 206 (18) (2005) 1847–1852.
[41] W. Lu, B. Ding, W. Zou, S. Dai, Direct synthesis of polyethylene thermoplastic elastomers with high molecular weight and excellent elastic recovery via a hybrid steric bulky strategy, Eur. Polym. J. 201 (2023) 112577.
[42] X. Hu, X. Kang, Y. Zhang, Z. Jian, Facile access to polar functionalized ultrahigh molecular weight polyethylene at ambient conditions, CCS Chem. 4 (5) (2022) 1680–1694.
[43] N. Nie, Y. Wang, C. Tan, M. Chen, D. Peng, The synthesis of hyperbranched ethylene oligomers by nickel catalysts with oxazole structure, J. Catal. 428 (2023) 115164.
[44] H. Deng, H. Zheng, C. Cheung, D. Li, H. Gao, Q. Lv, H. Gao, Synthesis of polyolefin elastomers from amine-imine nickel-catalyzed ethylene polymerization, Eur. Polym. J. 202 (2024) 112626.
[45] Z. Lu, X. Xu, Y. Luo, S. He, W. Fan, S. Dai, The unexpected effect of catalyst's structural symmetry on branching microstructure of polyethylene in late transition metal polymerization catalysis, ACS Catal. 13 (1) (2023) 725–734.
[46] W. Yuan, W. Li, S. Dai, Preparation of polyethylene thermoplastic elastomers using CS-symmetric nickel catalysts in ethylene polymerization, Polymer 285 (2023) 126322.
[47] W. Wang, Q. Wang, C. Zou, C. Chen, Synthesis of ultra-high-molecular-weight polyethylene by transition-metal-catalyzed precipitation polymerization, Precis. Chem. 2 (2) (2024) 63–69.
[48] J.R. Severn, J.C. Chadwick, V.V. Castelli, $MgCl_2$-based supports for the immobilization and activation of nickel diimine catalysts for polymerization of ethylene, Macromolecules 37 (17) (2004) 6258–6259.
[49] Z. Bi, J. Zhou, N. Zhu, D. Peng, C. Chen, Heterogeneous nickel catalysts for the synthesis of ethylene-based polyolefin elastomers, Macromolecules 57 (3) (2024) 1080–1086.

[50] D. Peng, C. Chen, Photoresponsive palladium and nickel catalysts for ethylene polymerization and copolymerization, Angew. Chem. Int. Ed. 113 (41) (2021) 22369–22374.
[51] K. Lian, Y. Zhu, W. Li, S. Dai, C. Chen, Direct synthesis of thermoplastic polyolefin elastomers from nickel catalyzed ethylene polymerization, Macromolecules 50 (16) (2017) 6074–6080.
[52] J. Li, D. Peng, C. Tan, C. Chen, Outer-shell self-supported nickel catalysts for the synthesis of polyolefin composites, Angew. Chem. Int. Ed. 62 (18) (2023) e202300359.